Electricity Demand Forecasting and Model Evaluation

SYNTHETIC EDUCATIONAL DOCUMENT — NOT AN OFFICIAL SOURCE

Purpose: Provide background material for retrieval and chatbot testing in the SmartGrid AI prototype.

This synthetic educational guide describes common load forecasting methods and evaluation practices. It does not provide validated forecasts for Texas or any utility.

1. Forecasting horizons

Very short-term forecasts support operational decisions over minutes or hours. Day-ahead forecasts estimate demand for the next day. Medium- and long-term forecasts support planning and investment. Each horizon requires different features and evaluation methods.

2. Explanatory variables

Common predictors include historical load, temperature, humidity, calendar indicators, hour, weekday, holidays, season, economic activity, and weather forecasts. Features must be available at the time the prediction is made to avoid data leakage.

3. Regression methods

Linear regression provides a simple baseline and interpretable coefficients. Random forests and gradient-boosted trees can capture nonlinear relationships and interactions. More complex models should be compared with simple baselines rather than assumed to be better.

4. Train and test design

For time-dependent data, chronological splitting is generally more realistic than randomly mixing past and future observations. A model should be evaluated on a period that was not used for fitting or tuning.

5. Metrics

MAE measures average absolute prediction error in the target's units. RMSE penalizes larger errors more strongly. R-squared describes variance explained relative to a baseline, but can be negative on test data. Metrics should be interpreted alongside the scale and distribution of demand.

6. Classification of grid stress

A classifier may estimate whether a defined stress condition is likely. The target label must have a clear operational definition. Accuracy alone can be misleading when high-stress cases are rare; precision, recall, F1 score, confusion matrix, and calibration are useful.

7. Limitations

Synthetic training data can help demonstrate a pipeline but cannot establish real-world predictive performance. A deployment requires representative historical observations, careful label design, validation across seasons and regions, and ongoing monitoring.

End of synthetic document. Verify operational facts with current official sources before real-world use.